INTEGRAL SPACES ARE DIRECTED LIMITS OF FINITE PRESENTED ONES, FIXED OVER V

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 3828-3908, following the statement at 3817-3826.

Editorial note. No omissions flagged: the 'similar to the proof of' and 'see proof of' pointers accompany fully written-out steps in this span.

Allowed standard prerequisites (editorial).

- Scheme finite over the base as a refinement base. proposition-there-is-a-scheme-finite-over (same chapter, cited at ~3829-3830). The lemma refines this proposition; same algebra strategy.
- Colimit presentation by finitely presented modules. lemma-colimit-finitely-presented (same chapter, cited at ~3846-3848). Writes A = colim F_i with F_i finitely presented O_Y-modules.
- Relative Spec for quasi-coherent algebras. Morphisms of Spaces, Definition spaces-morphisms-definition-relative-spec (cross-chapter, cited at ~3836-3838). X_i = relative Spec_Y(A_i).
- Colimit compatibility of finitely presented modules. Cohomology of Spaces, Lemma spaces-cohomology-lemma-finite-presentation-quasi-compact-colimit (cross-chapter, cited at ~3884-3885). Produces the compatible maps E_ik -> E_{i'k'}.
- Finite presented colimit of finite algebras. lemma-finite-algebra-directed-colimit-finite-finitely-presented and lemma-integral-algebra-directed-colimit-finite (same chapter, cited at ~3843-3845, ~3897-3898, ~3904-3905). A_ik finite and of finite presentation for k large; the first colimit equality.

Source reference. Original lemma 07V9: sources/stacks-spaces-limits/pinned-original-source.tex, source lines 2492–2492. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 086Z: sources/stacks-spaces-limits/pinned-original-source.tex, source lines 2639–2639. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 082C: sources/stacks-spaces-limits/pinned-original-source.tex, source lines 2679–2679. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has

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to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original proposition 09YC: sources/stacks-spaces-limits/pinned-original-source.tex, source lines 3786–3786. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 07U7: focused-seeds/geometry/S9-spaces-limits-integral-limit-finite-and-finite-presentation-refined/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 081W: focused-seeds/geometry/S9-spaces-limits-integral-limit-finite-and-finite-presentation-refined/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let S be a scheme. Let $f : X \to Y$ be an integral morphism of algebraic spaces over S. Assume Y quasi-compact and quasi-separated. Let $V \subset Y$ be a quasi-compact open subspace such that $f^{-1}(V) \to V$ is finite and of finite presentation. Then X can be written as a directed limit $X = \lim X_i$ where $f_i : X_i \to Y$ are finite and of finite presentation such that $f^{-1}(V) \to f_i^{-1}(V)$ is an isomorphism for all i.*

Proof. This lemma is a slight refinement of Proposition 09YC. Consider the integral quasi-coherent $\mathcal{O}_Y$-algebra $\mathcal{A} = f_*\mathcal{O}_X$. In the next paragraph, we will write $\mathcal{A} = \operatorname{colim} \mathcal{A}_i$ as a directed colimit of finite and finitely presented $\mathcal{O}_Y$-algebras $\mathcal{A}_i$ such that $\mathcal{A}_i|_V = \mathcal{A}|_V$. Having done this we set $X_i = \underline{\operatorname{Spec}}_Y(\mathcal{A}_i)$, see Morphisms of Spaces, Definition 081W. By construction $X_i \to Y$ is finite and of finite presentation, $X = \lim X_i$, and $f_i^{-1}(V) = f^{-1}(V)$.

The proof of the assertion on algebras is similar to the proof of part (2) of Lemma 082C. First, write $\mathcal{A} = \operatorname{colim} \mathcal{F}_i$ as a colimit of finitely presented $\mathcal{O}_Y$-modules using Lemma 07V9. Since $\mathcal{A}|_V$ is a finite type $\mathcal{O}_V$-module we may and do assume that $\mathcal{F}_i|_V \to \mathcal{A}|_V$ is surjective for all i. For each i, let $\mathcal{J}_i$ be the kernel of the map

$$\operatorname{Sym}^*_{\mathcal{O}_X}(\mathcal{F}_i) \longrightarrow \mathcal{A}$$

For $i' \geq i$ there is an induced map $\mathcal{J}_i \to \mathcal{J}_{i'}$. We have $\mathcal{A} = \operatorname{colim} \operatorname{Sym}^*_{\mathcal{O}_X}(\mathcal{F}_i)/\mathcal{J}_i$. Moreover, the quasi-coherent $\mathcal{O}_X$-algebras $\operatorname{Sym}^*_{\mathcal{O}_X}(\mathcal{F}_i)/\mathcal{J}_i$ are finite (as finite type quasi-coherent subalgebras of the integral quasi-coherent $\mathcal{O}_Y$-algebra $\mathcal{A}$ over $\mathcal{O}_X$). The restriction of $\operatorname{Sym}^*_{\mathcal{O}_X}(\mathcal{F}_i)/\mathcal{J}_i$ to V is $\mathcal{A}|_V$ by the surjectivity above. Hence $\mathcal{J}_i|_V$ is finitely generated as an ideal sheaf of $\operatorname{Sym}^*_{\mathcal{O}_X}(\mathcal{F}_i)|_V$ due to the fact that $\mathcal{A}|_V$ is finitely presented as an $\mathcal{O}_Y$-algebra. Write $\mathcal{J}_i = \operatorname{colim} \mathcal{E}_{ik}$ as a colimit of finitely presented $\mathcal{O}_X$-modules. We may and do assume that $\mathcal{E}_{ik}|_V$ generates $\mathcal{J}_i|_V$ as a sheaf of ideal of $\operatorname{Sym}^*_{\mathcal{O}_X}(\mathcal{F}_i)|_V$ by the statement on finite generation above.

Given $i' \geq i$ and k there exists a k' such that we have a map $\mathcal{E}_{ik} \to \mathcal{E}_{i'k'}$ making

$$\begin{array}{ccc} \mathcal{J}_i & \longrightarrow & \mathcal{J}_{i'} \\ \uparrow & & \uparrow \\ \mathcal{E}_{ik} & \longrightarrow & \mathcal{E}_{i'k'} \end{array}$$

commute. This follows from Cohomology of Spaces, Lemma 07U7. This induces a map

$$\mathcal{A}_{ik} = \mathrm{Sym}^*_{\mathcal{O}_X}(\mathcal{F}_i)/(\mathcal{E}_{ik}) \longrightarrow \mathrm{Sym}^*_{\mathcal{O}_X}(\mathcal{F}_{i'})/(\mathcal{E}_{i'k'}) = \mathcal{A}_{i'k'}$$

where $(\mathcal{E}_{ik})$ denotes the ideal generated by $\mathcal{E}_{ik}$. The quasi-coherent $\mathcal{O}_X$-algebras $\mathcal{A}_{ki}$ are of finite presentation and finite for k large enough (see proof of Lemma 086Z). Moreover we have $\mathcal{A}_{ik}|_V = \mathcal{A}|_V$ by construction. Finally, we have

$$\mathrm{colim}\, \mathcal{A}_{ik} = \mathrm{colim}\, \mathcal{A}_i = \mathcal{A}$$

Namely, the first equality was shown in the proof of Lemma 086Z and the second equality because $\mathcal{A}$ is the colimit of the modules $\mathcal{F}_i$. $\square$